239798: porphyritic dyke, Pannawonica Road

(Intrusion in Wittenoom Formation, Hamersley Group, Hamersley Basin, Capricorn Orogen)

Location and sampling

Yarraloola (SF 50-6), Pannawonica (2154)
MGA Zone 50, 418326E 7606614N

Sampled on 26 July 2019

This sample was collected from outcrop on Yalleen Station, about 16.4 km east-northeast of White Gate Well, 11.8 km west of Pannawonica townsite, and 2.6 km north of Pannawonica Road.

Geological context

The unit sampled is a porphyritic dyke intruded into the 2596–2506 Ma Wittenoom Formation of the Hamersley Group (Trendall et al., 2004; Martin, 2020). The sample locality is at the northwestern end of a northwest-trending, irregular, linear body of porphyritic rock that intrudes the upper Fortescue and lower Hamersley Groups to the west of Pannawonica (Williams et al., 1968). An interpreted extension of the intrusion about 8.0 km to the north-northwest, is within interpreted Turee Creek Group, and a similar east-trending intrusion about 25 km to the north cuts upper Fortescue Group rocks (Williams et al., 1968). About 17.9 km to the west-southwest, a similar unit with sill-like geometry intrudes the 2008–1799 Ma Duck Creek Dolomite of the Wyloo Group, and has an igneous crystallization age of 1781 ± 8 Ma (GSWA 225797, Wingate et al., 2021). Dacitic rocks about 73 km to the south-southwest have been dated at 1790–1787 Ma and tentatively assigned to the Capricorn Group (Evans et al., 2003; GSWA 219588, Wingate et al., 2019). Other rocks of similar age farther south include granitic rocks of the 1820–1775 Ma Moorarie Supersuite of the Gascoyne Province (Sheppard et al., 2020).

Petrographic description

The sample is a porphyritic pumice breccia of rhyolitic composition (Fig. 1), and consists of about 85% pumice clasts, 12% plagioclase, 1–2% quartz, 1–2% magnetite and accessory chlorite, apatite, pyrite, and titanium minerals. Pumice clasts are up to 15 mm in size and most are elongate and blocky in shape. Clast margins are generally poorly defined and the clasts are mainly identified by abrupt changes in the orientation of compacted tube vesicles. Most of the pumice is aphyric, although some contain phenocrysts of quartz and plagioclase up to 1 mm in size. Plagioclase phenocrysts are subhedral to euhedral, up to 5 mm in size, and mainly albitized. Most are not obviously within pumice clasts, although some may be, but mainly occur in aggregates. Magnetite is anhedral, embayed, up to 0.7 mm in size, and contains extremely fine probable titanium minerals and trace pyrite specks. Apatite is euhedral, tabular, up to 0.15 mm in size, and associated with some of the plagioclase aggregates. Although volcanic textures are clear, they have been overprinted by a moderately intense alteration assemblage, which consists of fine albite and quartz after pumice, with fine-grained chlorite enhancing the flattened vesicular texture. The pumice clasts have been compacted but the rock is not welded. The rock has not been deformed and metamorphic grade is very low.

Zircon morphology

Zircons isolated from this sample are colourless to dark brown and mostly subhedral to euhedral. The crystals are up to 300 µm long, and elongate, with aspect ratios up to 5:1. In cathodoluminescence (CL) images, concentric zoning is ubiquitous, and some zircons appear to contain older cores. A CL image of representative zircons is shown in Figure 2.

Analytical details

This sample was analysed on 25–26 September 2020, using SHRIMP-B. Eleven analyses of the M257 standard obtained during the session indicated an external spot-to-spot (reproducibility) uncertainty of 0.50% (1σ) and a $^{238}U/^{206}Pb^{*}$ calibration uncertainty of 0.21% (1σ). Calibration uncertainties are included in the errors of $^{238}U/^{206}Pb^{*}$ ratios and dates listed in Table 1. Common-Pb corrections were applied to all analyses using contemporaneous isotopic compositions determined according to the model of Stacey and Kramers (1975).

Results

Twenty analyses were obtained from 20 zircons. Results are listed in Table 1, and shown in a concordia diagram (Fig. 3).

Interpretation

The analyses are concordant to moderately discordant (Fig. 3). Two analyses are >5% discordant. The dates obtained from these two analyses (Group D; Table 1) are unreliable, and are considered not to be geologically significant. The remaining 18 analyses can be divided into two groups, based on their $^{207}Pb^{*}/^{206}Pb^{*}$ ratios.

Figure 1. Outcrop image for sample 239798: porphyritic dyke, Pannawonica Road

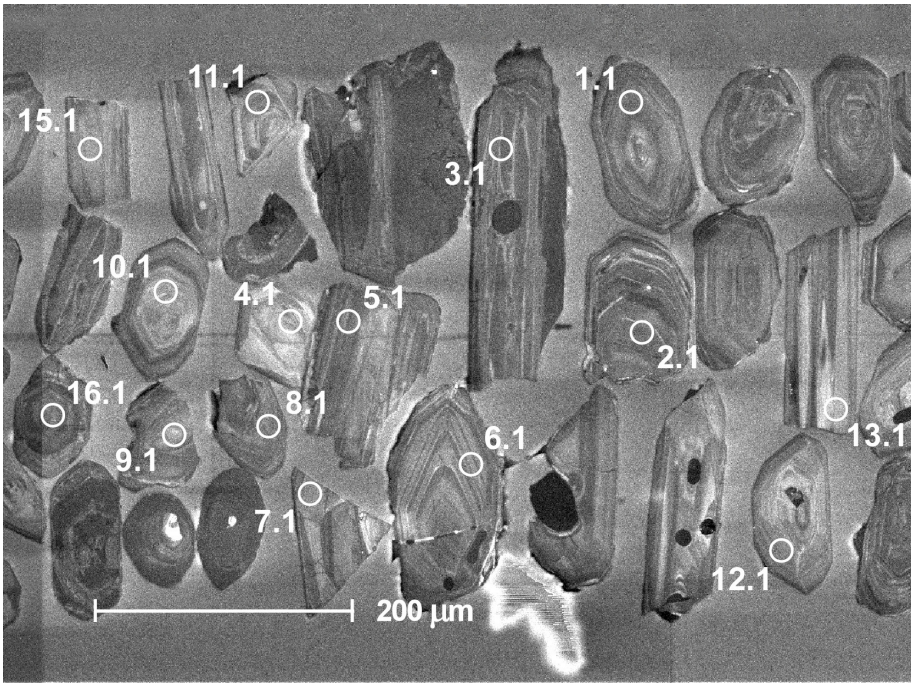

Figure 2. Cathodoluminescence image of representative zircons from sample 239798: porphyritic dyke, Pannawonica Road. Numbered circles indicate the approximate locations of analysis sites

Table 1. Ion microprobe analytical results for zircons from sample 239798: porphyritic dyke, Pannawonica Road

Group ID	Spot no.	Grain. spot	^{238}U (ppm)	^{232}Th (ppm)	$^{232}Th/^{238}U$	f204 (%)	$^{238}U/^{206}Pb$	± 1σ	$^{207}Pb/^{206}Pb$	± 1σ	$^{238}U/^{206}Pb^*$	± 1σ	$^{207}Pb^*/^{206}Pb^*$	± 1σ	$^{238}U/^{206}Pb^*$ date (Ma)	±1σ	$^{207}Pb^*/^{206}Pb^*$ date (Ma)	±1σ	Disc. (%)
I	20	20.1	212	96	0.45	-0.021	3.192	0.030	0.10867	0.00082	3.191	0.030	0.10885	0.00084	1757	15	1780	14	1.3
I	10	10.1	200	78	0.39	0.042	3.178	0.030	0.10932	0.00083	3.179	0.030	0.10895	0.00087	1763	15	1782	15	1.1
I	12	12.1	213	106	0.50	0.000	3.192	0.030	0.10969	0.00083	3.192	0.030	0.10969	0.00083	1757	15	1794	14	2.1
I	7	7.1	178	71	0.40	0.024	3.134	0.031	0.11003	0.00088	3.134	0.031	0.10982	0.00091	1785	16	1796	15	0.6
I	13	13.1	133	46	0.34	-0.065	3.185	0.035	0.10927	0.00103	3.183	0.035	0.10984	0.00110	1761	17	1797	18	2.0
I	14	14.1	179	60	0.33	0.052	3.164	0.032	0.11040	0.00092	3.166	0.032	0.10995	0.00097	1770	16	1799	16	1.6
I	4	4.1	181	87	0.48	0.000	3.102	0.031	0.11045	0.00086	3.102	0.031	0.11045	0.00086	1801	16	1807	14	0.3
I	17	17.1	212	88	0.42	-0.115	3.164	0.030	0.10944	0.00086	3.160	0.030	0.11045	0.00097	1772	15	1807	16	1.9
I	11	11.1	245	62	0.25	0.000	3.167	0.028	0.11051	0.00072	3.167	0.028	0.11051	0.00072	1769	14	1808	12	2.2
I	18	18.1	189	76	0.40	0.000	3.069	0.031	0.11077	0.00091	3.069	0.031	0.11077	0.00091	1818	16	1812	15	-0.3
X	3	3.1	581	322	0.55	0.000	2.305	0.047	0.15706	0.00054	2.305	0.047	0.15706	0.00054	2323	40	2424	6	4.2
X	9	9.1	85	39	0.45	0.161	2.243	0.043	0.15927	0.00136	2.246	0.043	0.15783	0.00150	2374	38	2433	16	2.4
X	8	8.1	234	88	0.37	0.173	2.259	0.021	0.16002	0.00085	2.263	0.021	0.15848	0.00095	2359	18	2440	10	3.3
X	19	19.1	276	108	0.39	-0.011	2.195	0.020	0.15938	0.00079	2.195	0.020	0.15948	0.00079	2420	18	2450	8	1.2
X	1	1.1	418	191	0.46	0.084	2.226	0.029	0.16030	0.00063	2.228	0.029	0.15955	0.00066	2390	26	2451	7	2.5
X	6	6.1	233	60	0.26	-0.012	2.146	0.020	0.15967	0.00083	2.146	0.020	0.15977	0.00084	2466	19	2453	9	-0.5
X	16	16.1	290	111	0.38	-0.011	2.194	0.020	0.15970	0.00080	2.194	0.020	0.15980	0.00080	2421	19	2454	9	1.3
X	2	2.1	311	174	0.56	0.113	2.159	0.019	0.16372	0.00073	2.162	0.019	0.16271	0.00079	2451	18	2484	8	1.3
D	15	15.1	202	42	0.21	-0.067	3.293	0.031	0.11312	0.00088	3.291	0.031	0.11371	0.00094	1710	14	1860	15	8.0
D	5	5.1	323	233	0.72	0.185	2.275	0.019	0.16602	0.00073	2.280	0.019	0.16437	0.00081	2345	17	2501	8	6.3

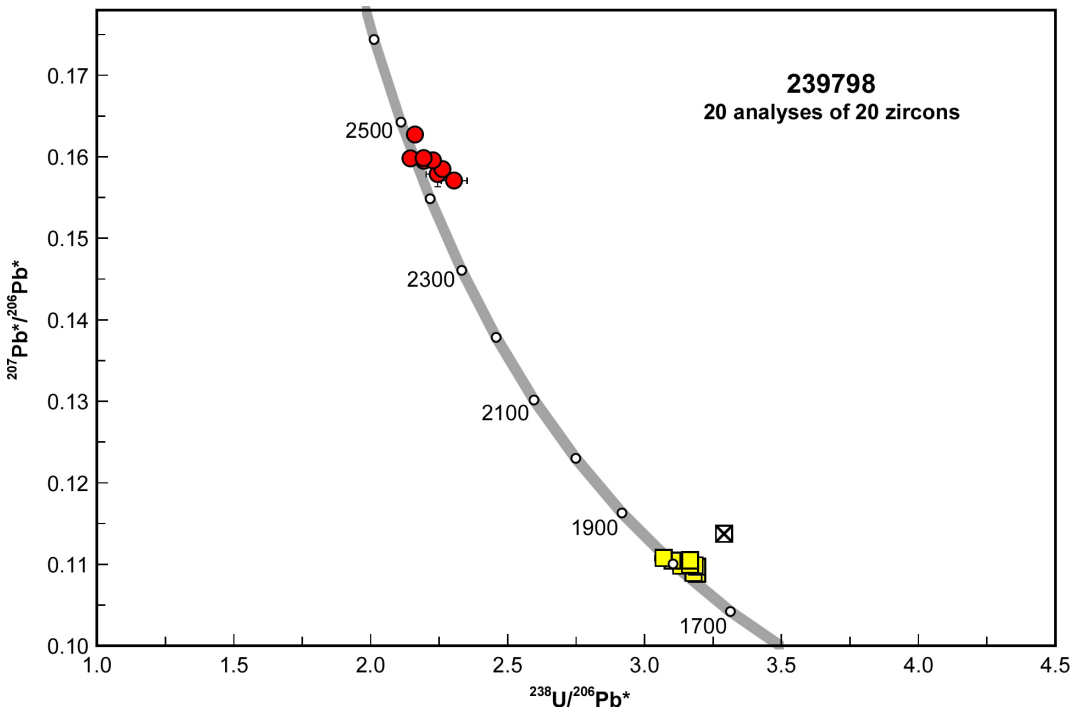

Figure 3. U–Pb analytical data for zircons from sample 239798: porphyritic dyke, Pannawonica Road. Yellow squares indicate Group I (magmatic zircons); red circles indicate Group X (xenocrystic zircons); crossed squares indicate Group D (discordance >5%)

Group I comprises 10 analyses (Table 1), which yield a weighted mean $^{207}Pb^*/^{206}Pb^*$ date of 1798 ± 9 Ma (MSWD = 0.57).

Group X comprises eight analyses (Table 1), which yield $^{207}Pb^*/^{206}Pb^*$ dates of 2484–2424 Ma. Six analyses in Group X yield a weighted mean $^{207}Pb^*/^{206}Pb^*$ date of 2449 ± 7 Ma (MSWD = 0.50).

The date of 1798 ± 9 Ma for the 10 analyses in Group I is interpreted as the igneous crystallization age of the porphyritic dyke. The dates of 2484–2424 Ma for the eight analyses in Group X are interpreted as the ages of inherited zircons. The date of 2449 ± 7 Ma for six analyses in Group X is interpreted as the age of an inherited component.

References

Evans, DAD, Sircombe, KN, Wingate, MTD, Doyle, M, McCarthy, M, Pidgeon, RT and Van Niekerk, HS 2003, Revised geochronology of magmatism in the western Capricorn Orogen at 1805–1785 Ma: Diachroneity of the Pilbara–Yilgarn collision: Australian Journal of Earth Sciences, v. 50, p. 853–864.

Martin, DMcB 2020, Wittenoom Formation (A-HAd-kd): Geological Survey of Western Australia, WA Geology Online, Explanatory Notes extract, viewed 10 September 2021, <www.dmp.wa.gov.au/ens>.

Sheppard, S, Johnson, SP and Occhipinti, SA 2020, Moorarie Supersuite (P_-MO-g): Geological Survey of Western Australia, WA Geology Online, Explanatory Notes extract, viewed 07 September 2021, <www.dmp.wa.gov.au/ens>.

Stacey, JS and Kramers, JD 1975, Approximation of terrestrial lead isotope evolution by a two-stage model: Earth and Planetary Science Letters, v. 26, p. 207–221.

Trendall, AF, Compston, W, Nelson, DR, de Laeter, JR and Bennett, VC 2004, SHRIMP zircon ages constraining the depositional chronology of the Hamersley Group, Western Australia: Australian Journal of Earth Sciences, v. 51, no. 5, p. 621–644.

Williams, IR, Ryan, GR and Halligan, R 1968, Yarraloola, WA Sheet SF 50-6: Geological Survey of Western Australia, 1:250 000 Geological Series.

Wingate, MTD, Lu, Y, Fielding, IOH and Johnson, SP 2019, 219588: porphyritic dacite, One Tree Well; Geochronology Record 1570: Geological Survey of Western Australia, 4p.

Wingate, MTD, Fielding, IOH, Lu, Y and Martin, DMcB 2021, 225797: porphyritic microgranite, Three Peak Hill; Geochronology Record 1814: Geological Survey of Western Australia, 4p.

Recommended reference for this publication

Wingate, MTD, Fielding, IOH, Lu, Y and Martin, DMcB 2021, 239798: porphyritic dyke, Pannawonica Road; Geochronology Record 1818: Geological Survey of Western Australia, 4p.

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